

Senior Project: Network Design

1. Network Design Overview

This document covers **only the Network Design portion** of the Senior Project. It defines the logical topology, VLAN segmentation, IP addressing, routing, and perimeter security placement.

This network design serves as the **foundation** for later project phases, including security monitoring, attack simulation, detection, and analysis.

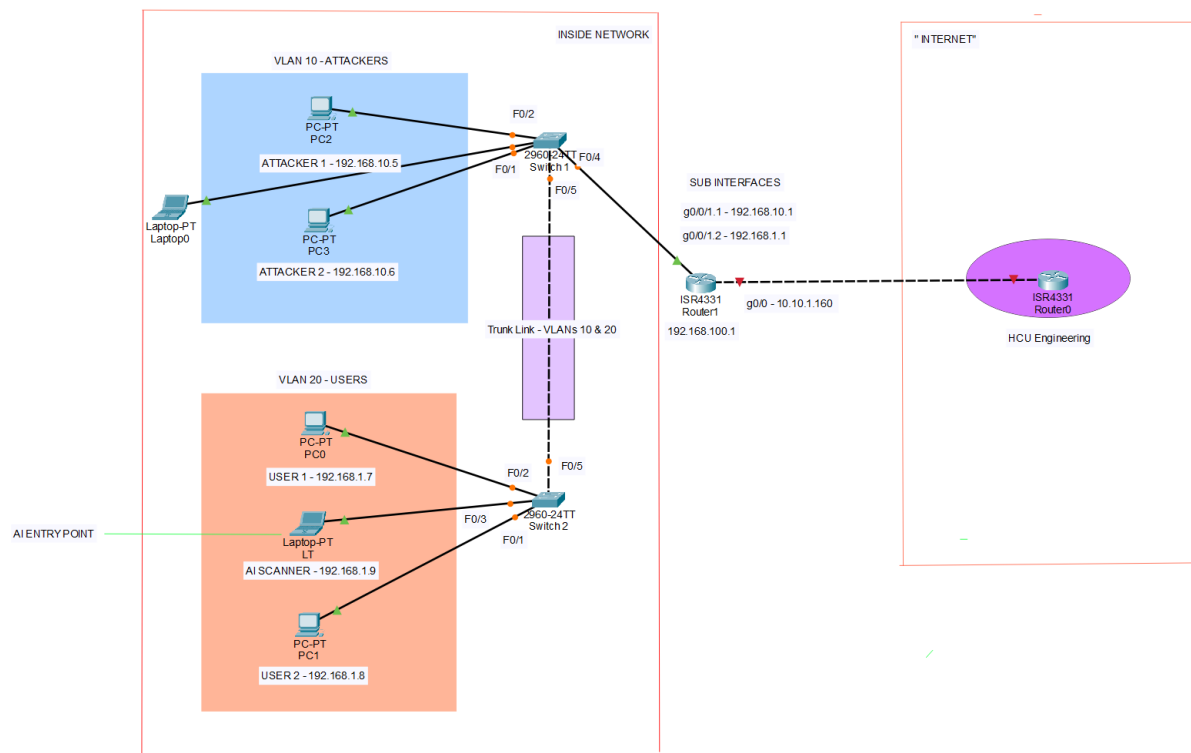
2. Network Design Objectives

- Implement VLAN-based network segmentation
- Enable inter-VLAN routing using a router
- Simulate attacker and user networks
- Enforce perimeter security using a firewall
- Understand packet flow between security zones

3. Network Topology Summary

Figure 1: Network Topology Diagram

The following diagram illustrates the complete network design, including VLAN segmentation, inter-switch trunking, router-on-a-stick configuration, firewall placement, and the simulated internet connection.



The network design uses **two Layer 2 access switches** connected via a **trunk link** to aggregate VLAN traffic before routing. A single router interface is used for **router-on-a-stick (ROAS)** inter-VLAN routing, conserving physical ports and enabling firewall connectivity.

Logical Zones

- **Inside Network:** VLAN 10 (Attackers) and VLAN 20 (Users)
- **Perimeter Network:** Point-to-point link between Router and Firewall
- **Outside Network:** Simulated Internet router

VLAN Definitions

VLAN 10 – ATTACKERS

- Purpose: Simulated attacker machines
- Network: 192.168.10.0/24

- Default Gateway: 192.168.10.1

VLAN 20 – USERS

- Purpose: Internal user machines
 - Network: 192.168.1.0/24
 - Default Gateway: 192.168.1.1
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4. Devices Used

End Devices

- PC2 – Attacker 1 (192.168.10.5)
- PC3 – Attacker 2 (192.168.10.6)
- PC0 – User 1 (192.168.1.7)
- PC1 – User 2 (192.168.1.8)

Network Devices

- 2 × Cisco 2960-24TT Layer 2 Switches
 - 1 × Cisco ISR4331 Router (Internal Routing)
 - 1 × Cisco ASA 5506-X Firewall
 - 1 × Cisco ISR4331 Router (Internet Simulation)
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5. IP Addressing Scheme

Attacker VLAN (VLAN 10)

- Network: 192.168.10.0/24
- Router Interface: g0/0/1 – 192.168.10.1

User VLAN (VLAN 20)

- Network: 192.168.1.0/24
 - Router Interface: g0/0/0 – 192.168.1.1
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6. DHCP Configuration

The router provides dynamic IP address assignment for both VLANs.

DHCP Pool – VLAN 10 (ATTACKERS)

```
ip dhcp excluded-address 192.168.10.1
```

```
ip dhcp pool VLAN10  
network 192.168.10.0 255.255.255.0  
default-router 192.168.10.1  
dns-server 8.8.8.8
```

DHCP Pool – VLAN 20 (USERS)

```
ip dhcp excluded-address 192.168.1.1
```

```
ip dhcp pool VLAN20  
network 192.168.1.0 255.255.255.0  
default-router 192.168.1.1  
dns-server 8.8.8.8
```

DHCP Design Explanation

- The router assigns IP addresses dynamically
- The gateway address is excluded from the pool
- DNS server is configured automatically
- End devices are set to obtain IP configuration automatically

This design reflects real-world enterprise deployment.

7. Switch Configuration Overview

VLAN Configuration (Both Switches)

Both switches have **VLAN 10 and VLAN 20 created**, even if end devices from a VLAN are not directly connected. This ensures VLAN traffic can traverse the trunk link.

- VLAN 10 – ATTACKERS
- VLAN 20 – USERS

Access Ports

- Switch 1 access ports are assigned to **VLAN 10 (Attackers)**
- Switch 2 access ports are assigned to **VLAN 20 (Users)**

Inter-Switch Trunk Link

The switches are connected using an **802.1Q trunk link** that carries VLANs 10 and 20.

- Trunk Interface: F0/5 (both switches)
- Allowed VLANs: 10, 20

Router Uplink

- Switch 1 connects to the router using a **trunk port** to support router-on-a-stick routing

8. Router Configuration Overview

The internal router uses **router-on-a-stick (ROAS)** to perform inter-VLAN routing for VLAN 10 and VLAN 20 over a single physical interface.

Router Subinterfaces

Subinterface	VLAN	IP Address	Purpose
g0/0/1.1	10	192.168.10.1	Attacker VLAN Gateway
g0/0/1.2	20	192.168.1.1	User VLAN Gateway
g0/0/0		10.10.1.160	Connects to HCU Engineering

9. Traffic Flow Explanation

Inter-VLAN Communication

1. A device sends traffic to a destination outside its subnet
2. The packet is forwarded to its default gateway

3. The router routes the packet to the destination VLAN
4. The destination switch delivers the packet to the target device

Internet-Bound Traffic

1. Internal host sends traffic to the default gateway
 2. Router forwards traffic to the ASA firewall
 3. Firewall applies security policies
 4. Allowed traffic is forwarded to the internet router
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10. Security Design Considerations

- VLAN segmentation reduces lateral movement
 - Attacker network is isolated from user network
 - Router provides controlled inter-VLAN routing
 - Architecture supports future IDS/IPS integration
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11. Testing and Validation

Tests Performed

- Ping within same VLAN (success)
- Ping between VLANs (success via router)
- Internet-bound traffic path verification

Expected Results

- VLAN isolation at Layer 2
 - Controlled routing at Layer 3
 - Centralized security enforcement at the firewall
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13. Network Design Summary

This network design establishes a segmented, routable, and securable infrastructure that mirrors real-world enterprise environments. By separating attacker systems, internal users, and external networks, the design provides a controlled platform for implementing security controls, traffic monitoring, and attack detection in later project stages.

14. Next Project Phases (Beyond Network Design)

- Implement ACLs between VLANs
- Enforce firewall NAT and filtering rules

Network Connectivity Verification

Overview

This report documents the successful configuration and verification of internet connectivity for the internal network consisting of VLAN 10 (Attackers) and VLAN 20 (Users). The network utilizes a router-on-a-stick configuration with NAT (Network Address Translation) to enable external communication through the HCU Engineering network.

Network Configuration Summary

- **Internal Interfaces:**
 - GigabitEthernet0/1.1 → 192.168.10.1 (VLAN 10)
 - GigabitEthernet0/1.2 → 192.168.1.1 (VLAN 20)
 - **External Interface:**
 - GigabitEthernet0/0 → 10.10.1.160 (assigned via DHCP)
 - **Default Route:**
 - 0.0.0.0/0 via 10.10.1.1
 - **NAT Configuration:**
 - Inside interfaces: g0/1.1, g0/1.2
 - Outside interface: g0/0
 - Access Control List:
 - 192.168.0.0/16 permitted
 - NAT Overload (PAT) enabled using g0/0
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Verification Steps and Results

1. Internal Connectivity

- Devices in VLAN 10 and VLAN 20 successfully communicate with each other.
- Devices can reach their respective default gateways.
- Router subinterfaces are operational and correctly configured.

Result: Successful

2. External Interface Connectivity

- Router interface g0/0 (10.10.1.160) is reachable.
- Router can successfully ping the upstream gateway (10.10.1.1) when sourcing traffic from g0/0.

Result: Successful

3. Routing Verification

- Default route is present in the routing table:
 - 0.0.0.0/0 via 10.10.1.1
- Connected routes for all internal and external networks are correctly installed.

Result: Successful

4. NAT Operation Verification

- NAT translations show internal IPs (192.168.x.x) being translated to 10.10.1.160.
- Example translation:
 - 192.168.10.8 → 10.10.1.160 → external internet addresses
- NAT statistics:
 - Active translations: 65
 - Hits: 3986
 - Misses: 0

Result: NAT is functioning correctly

5. Internet Connectivity

- Traffic is successfully reaching external IP addresses such as:
 - 8.8.8.8
 - Multiple public IPs (e.g., 52.x.x.x, 104.x.x.x)
- Return traffic is successfully received.

Result: Full internet connectivity achieved

Conclusion

The network is fully operational and properly configured. All required components for internet access are functioning as expected:

- Internal VLAN routing is operational.
- The router has a valid default route to the upstream network.
- NAT (PAT) is correctly translating internal private IP addresses to the external interface.
- Bidirectional communication with external networks is confirmed.

No further configuration changes are required.